Predicting Individual Residential Electricity Consumption

A Neural Network Modeling Approach

ECE 590 Neural Networks

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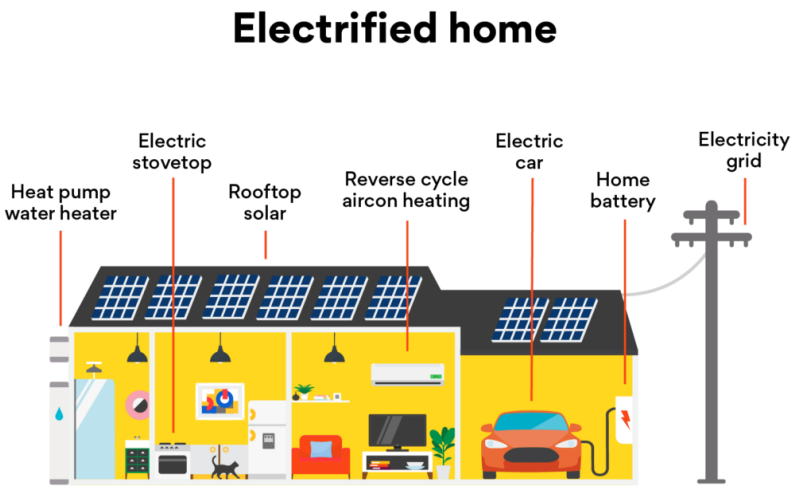
Problem Motivation

Residential heat pump adoption is accelerating as households electrify to reduce emissions and heating costs.

My residence is fully electrified and heated primarily with heat pumps, so the electric meter is a direct proxy for HVAC load plus occupant behavior.

Accurate prediction matters for equipment sizing, degradation detection, and retrofit decisions.



https://www.originenergy.com.au/blog/how-to-electrify-your-home/

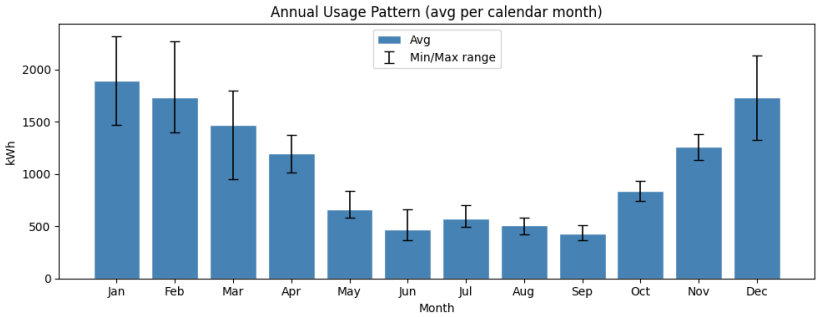
Simplest Solution

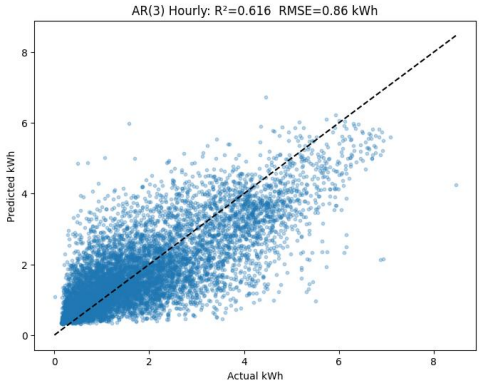
Autoregressive model:

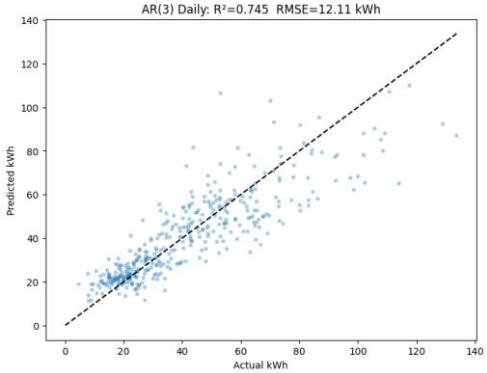
$$X_t = \sum_{i=1}^{p} \varphi_i X_{t-i} + \varepsilon_t$$

Predict the next value only from past values. Last hour's consumption drives this hour's consumption.

Consumption is highly seasonal at the monthly timescale.





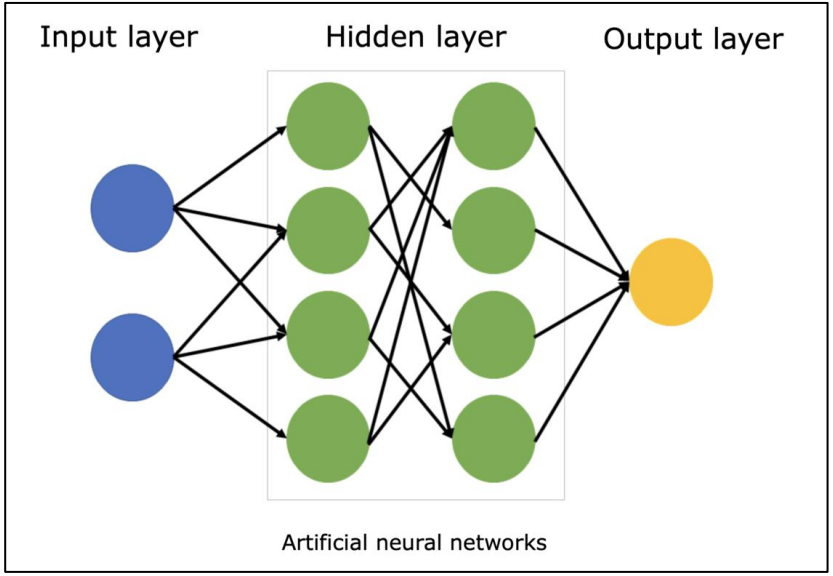


Why Neural Networks?

Autoregression requires utility smart-meter data that is not independently obtainable.

Factors driving consumption are non-linear phenomena (occupant behavior, weather, climate, seasonality, etc.)

Neural networks can find non-linear relationships in large datasets.



https://www.sciencelearn.org.nz/images/5156-neural-network-diagram

Data

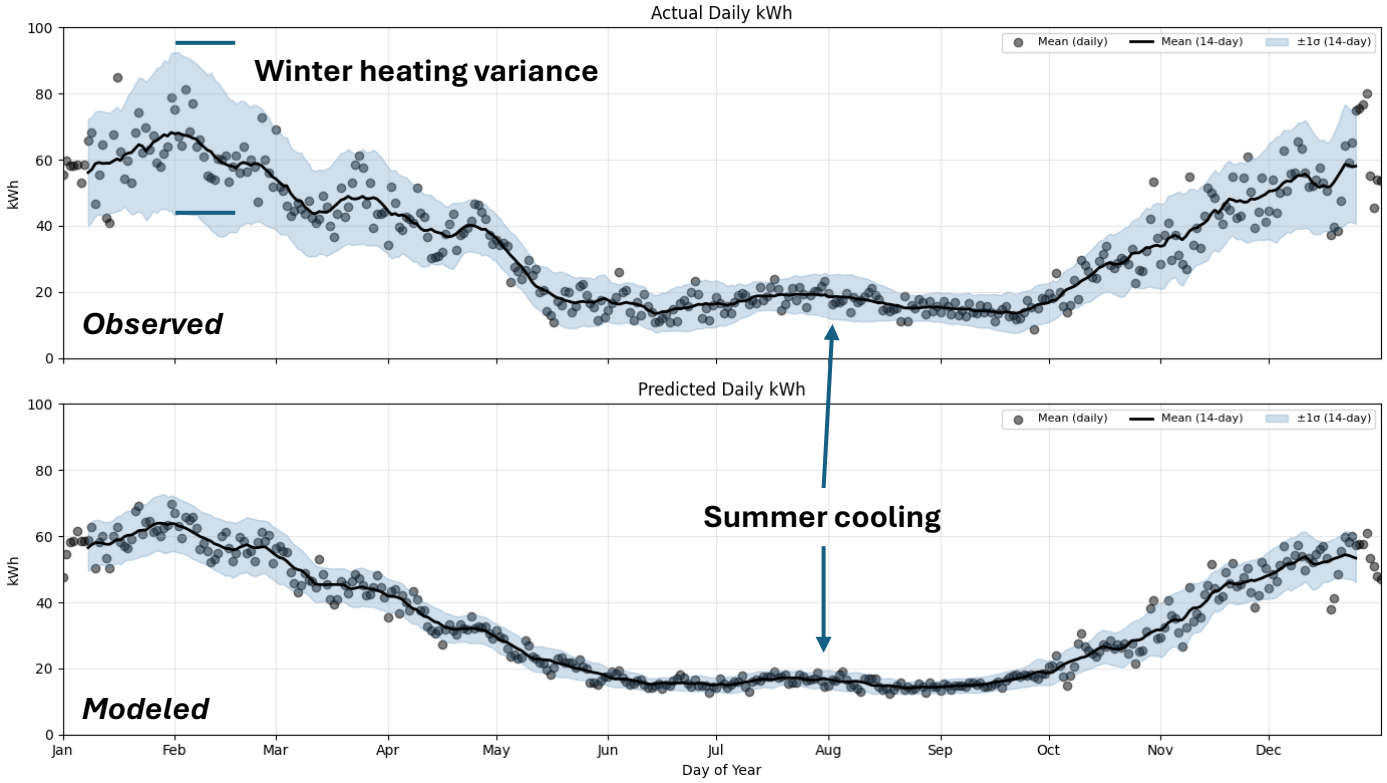
Category	Features	Description
Current Weather	temp, rhum, prcp, wspd	Real-time temperature, humidity, precipitation, and wind speed.
Wind Direction	wdir_sin, wdir_cos	Cyclical encoding of wind direction to maintain angular continuity.
Weather History	temp_lag1, lag3, lag6, temp_avg_12, rhum_lag3, prcp_sum_12	Historical trends and rolling averages (e.g., 12-hour total rain).
Human Activity	awake, evening	Binary flags for typical occupancy and activity periods.
Temporal Context	is_weekend, hour_sin/cos, month_sin/cos	Cyclical time of day and year to capture seasonal and weekly patterns.

Results

Timescale	Model	R^2	RMSE (kWh)
Hourly	Linear Reg.	0.384	1.10
Hourly	MLP	0.512	0.98
Hourly	LSTM	0.518	0.97
Hourly	**AR(3)**	**0.616**	**0.86**
Daily	Linear Reg.	0.678	13.59
Daily	MLP	0.719	12.69
Daily	LSTM	0.709	12.92
Daily	**AR(3)**	**0.745**	**12.11**

R^2 on temporal 75/25 test split.

The MLP captures nonlinear weather-to-load structure that linear regression misses. But three hours of past consumption beats every weather-feature model.

For single-household prediction, lagged consumption is a denser summary of household state than ambient weather. Improving prediction needs richer features, not deeper networks.

MLP Results Visualized